

AMENDMENT NO. 1 · OCTOBER 1979

TO

IS : 4382-1967 SPECIFICATION FOR A NON-TINTED OPHTHALMIC GLASS

Alterations

(Page 6, clause 4.4) — Substitute the following for the existing clause:

‘4.4 Ophthalmic glass shall be free from strain which causes double refraction. The process of annealing during manufacture, which is responsible for the presence of strain, shall be such that the strain is not more than 20 nm per centimetre of light path. The strain shall be symmetrically distributed. The strain shall be determined by the methods described in 6.2.’

(Page 6, clause 4.8) — Substitute the following for the existing clause:

‘4.8 When tested for the transmissivity of white light using a standard photometer, ophthalmic glass shall indicate absorption of not more than two percent (for crown glass) and three percent (for flint glass) of the incident light per centimetre of the light path.’

(Page 6, clause 4.9) — Delete the clause and renumber the subsequent clauses accordingly.

[Page 6, clause 4.12 (renumbered as 4.11), line 2] — Substitute ‘58 to 59’ for ‘58.2 to 58.9’.

[Page 6, clause 4.13 (renumbered as 4.12)] — Delete this clause and renumber the subsequent clauses accordingly:

[Page 7, clause 4.15 (renumbered as 4.13), lines 1 and 2] — Substitute ‘36 to 30’ for ‘35.7 to 30.3’.

[Page 7, clause 4.16 (renumbered as 4.14)] — Delete.

(Page 7, clause 6.2) — Substitute the following for the existing clause:

‘6.2 Test for Strain — Ophthalmic glass blanks shall be tested for strain by the methods described in 6.2.1 and 6.2.2. While Method A (see 6.2.1) is suitable for samples with heavy strain (over 100 nm), Method B is to
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be used where the total strain in the specimen is less than 100 nm. Method B is, therefore, the one to be generally employed particularly in the case of samples of optical glass, provision of testing facilities for Method A will also be desirable in order to do a preliminary screening by the use of Babinet Compensator, that is, to determine the order of magnitude of the strain and to determine which of the two extinction positions is the correct one.

6.2.1 Method A -- The optical bench arrangement for this test shall be as shown in Fig. 1.

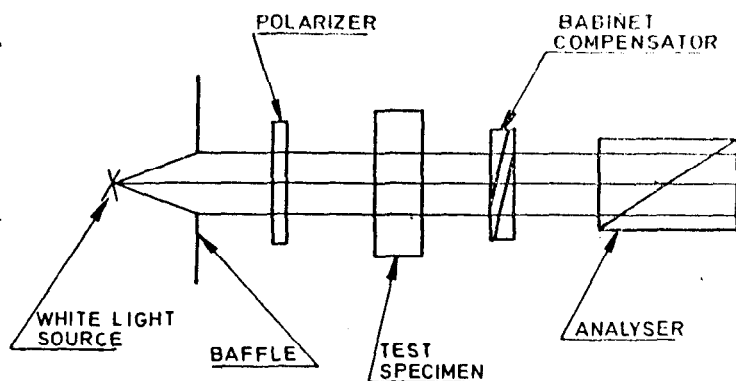


FIG. 1

Initially the polarizer and analyser are set in the crossed position and the compensator is set 45° to the polarizer when several coloured and one dark band is seen in the field. The compensator micrometer screw is moved till the dark band is under the reference line etched on the first prism. This shall be taken as micrometer reading R_1 . The distance between the initial position with the dark band under the etched reference line and next colour band is measured (*reading R_2*) on the micrometer screw. For more accurate measurement white light may be replaced by monochromatic light or a suitable colour filter may also be used.

The sample is introduced as shown in the diagram. The maximum displacement of the dark band is noted, rotating the sample when white light is used. If the displacement of the dark band is less than a quarter of the space between the dark band and the next coloured band, the set up given for Method B (see 6.2.2) shall be used for measurement of strain. If the displacement of the dark band is nearly a quarter or more, Babinet Compensator itself shall be used for the measurement of strain. (Sometimes the displacement may be several bands).

The total strain of the sample in nanometres is equal to the displacement of the micrometer from the initial position to, when the sample is introduced and rotated till the maximum displacement is observed, divided by the displacement for one fringe, multiplied by the wave length of the light used in nanometres.

For example if R_1 is the initial reading when the black fringe is under the etched line, R_2 the reading when the line is moved to the next fringe, R_3 the reading with the sample rotated to give maximum displacement and micrometer moved till the line is once again under the black line, then the strain is $\frac{R_3 - R_1}{R_2 - R_1} \times \lambda$. Strain for unit length is obtained by dividing the total strain by the length of the passage of the beam through the specimen.

If the strain is such that the displacement of the black fringe cannot be measured conveniently on the micrometer screw of the compensator, Method B given in 6.2.2 shall be used.

6.2.2 Method B — In the test set up shown in Fig. 2, the polarizer and analyser are set in crossed position.

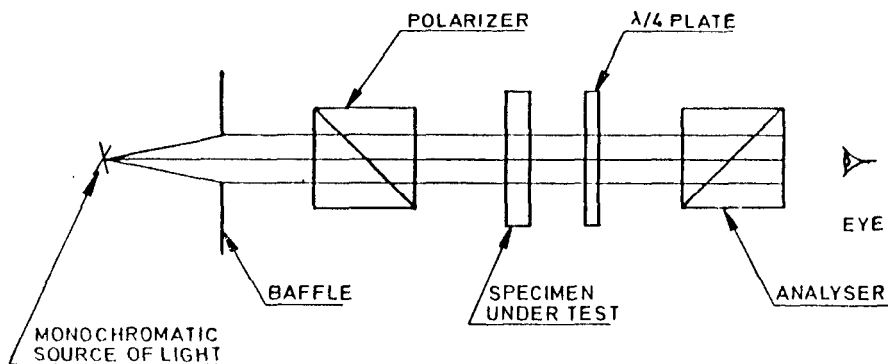


FIG. 2

With the quarter wave plate introduced and rotated till the light is extinguished, the specimen under test is introduced and rotated perpendicular to the optical axis of the system till the light is again extinguished. The specimen is rotated accurately 45° from the position of the extinction when maximum light is seen in the field. The analyser is rotated till the light is again extinguished. The rotation θ of the analyser is noted. The total strain in the specimen is given as $\theta \times \lambda/180$ where θ is the rotation of the analyser and λ is the wavelength of the light expressed in nanometres. For sodium light the strain for a rotation of angle θ expressed

in degrees is $3.27 \times \theta$ nm. The strain per unit length is obtained by dividing the total strain by the length of the path of the ray through the specimen. The following factors shall be borne in mind for this test:

- a) The quarter-wave plate shall be used with monochromatic light for which it is intended,
- b) The specimen stage and the analyser shall be provided with a facility of rotation to an accuracy of at least one degree,
- c) The extinction angle as obtained by this method will be θ or $(180 - \theta)$ but the initial examination by the Babinet Compensator will reveal the correct angle, and
- d) The percentage transmission in the crossed position between the polarizer and the analyser should not be more than 0.5 percent.

(EDC 36)